

SYDNEY HORSLEY PARK, Australia

WMO: 947600

Lat: 33.8511S

Lon: 150.8567E

Elev: 328

StdP: 14.52

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.2	40.6	22.8	17.4	67.4	26.7	20.9	63.2	20.8	62.2	17.9	59.5	1.8	310	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	95.8	68.2	91.2	68.3	87.1	67.4	73.6	85.7	72.1	82.7	70.9	80.5	9.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.7	114.3	76.5	69.4	109.2	75.2	68.1	104.6	74.1	37.3	85.6	36.0	82.9	35.0	80.5	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	16.6	14.3	33.1	105.5	1.8	4.7	31.8	108.8	30.7	111.6	29.7	114.2	28.4	117.7		
(5)			32.0	76.6	1.9	1.6	30.7	77.8	29.6	78.7	28.5	79.6	27.2	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	74.3	72.8	70.0	64.8	58.5	54.4	52.8	54.8	59.9	64.2	68.1	71.2
	DBStd	8.71	5.47	5.00	4.37	4.27	3.84	3.44	3.66	4.00	5.11	5.60	5.84	5.57
	HDD50	20	0	0	0	0	0	5	11	4	0	0	0	0
	HDD65	1580	1	2	7	53	204	319	378	318	172	85	31	10
	CDD50	5045	755	638	619	445	264	136	99	151	296	441	542	659
	CDD65	1127	291	220	161	48	2	0	0	1	17	61	123	203
	CDH74	8491	2273	1446	850	273	20	0	2	12	247	641	1097	1630
	CDH80	3280	1018	543	234	64	0	0	0	0	64	221	457	679
Wind	WSAvg	5.8	6.6	6.3	5.5	5.1	4.9	5.3	5.1	5.5	5.8	6.3	6.7	6.5
Precipitation	PrecAvg	35.4	4.2	4.1	4.1	3.0	2.8	2.6	1.6	2.1	1.8	2.8	3.5	3.0
	PrecMax	57.5	12.0	18.0	13.6	14.3	8.5	12.0	5.7	18.0	6.7	7.7	15.4	10.5
	PrecMin	20.5	0.2	0.3	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.4	0.1
	PrecStd	9.4	3.0	3.6	3.0	3.0	2.5	2.8	1.5	3.3	1.5	2.1	2.8	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.4	99.3	91.2	87.1	76.7	69.3	72.1	75.4	86.6	92.0	98.6	99.5
		MCWB	70.8	69.0	67.3	63.1	58.9	56.3	55.3	54.8	60.1	60.7	62.7	68.3
	2%	DB	95.6	91.5	85.7	80.3	73.0	65.9	66.9	71.0	80.9	85.7	91.2	92.8
		MCWB	70.7	71.3	69.0	63.2	58.5	55.0	52.3	54.0	57.4	60.5	65.2	67.9
	5%	DB	90.2	85.9	82.0	76.5	70.1	63.8	63.9	67.8	75.9	81.3	84.5	87.2
		MCWB	70.2	69.9	68.0	63.1	57.8	54.2	51.8	53.3	56.6	60.7	65.0	66.7
	10%	DB	85.3	82.2	78.5	73.5	67.4	61.8	61.4	64.8	71.4	76.4	80.0	82.8
		MCWB	69.3	68.6	66.8	62.4	56.5	53.5	51.0	51.8	55.6	59.9	64.2	66.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.2	75.6	73.5	68.9	64.4	60.9	59.9	60.2	63.6	67.9	71.7	73.6
		MCDB	91.0	88.9	81.9	77.6	68.9	65.0	65.9	68.4	74.2	78.5	86.1	87.1
	2%	WB	73.7	73.3	71.3	66.7	62.0	58.3	56.6	57.6	61.4	65.7	69.5	71.7
		MCDB	86.6	84.1	79.0	73.9	67.0	61.5	61.2	64.2	71.9	75.6	81.0	84.0
	5%	WB	72.3	71.7	69.8	65.4	60.1	56.9	54.5	55.6	59.6	64.0	67.7	70.0
		MCDB	83.5	81.0	77.3	71.9	66.3	60.5	59.5	63.9	69.6	73.6	78.2	80.6
	10%	WB	71.0	70.3	68.4	64.1	58.5	55.6	52.7	53.8	57.9	62.4	66.1	68.6
		MCDB	80.8	78.6	75.6	70.3	64.7	59.7	58.4	61.1	67.4	70.8	75.3	78.1
Mean Daily Temperature Range	5% DB	MDBR	19.5	16.9	16.7	17.5	18.9	16.1	18.7	20.5	21.7	21.0	19.8	19.8
		MCDBR	30.1	25.0	22.6	22.6	22.8	19.2	22.3	25.4	29.9	30.3	30.4	29.1
		MCWBR	9.1	7.9	8.5	8.9	10.5	10.5	11.3	11.7	11.5	10.8	9.9	9.3
	5% WB	MCDBR	23.3	20.1	19.2	18.4	18.2	13.8	16.6	20.1	23.3	22.8	23.2	23.1
		MCWBR	8.6	7.5	8.2	8.1	9.4	9.0	10.3	10.7	10.6	10.0	9.3	9.0
Clear-Sky Solar Irradiance	taub	0.391	0.389	0.359	0.331	0.315	0.306	0.296	0.303	0.335	0.356	0.381	0.391	
	taud	2.446	2.472	2.539	2.592	2.586	2.610	2.620	2.591	2.499	2.470	2.442	2.424	
	Ebn at Noon	300	294	292	284	272	266	276	289	295	301	301	301	
	Edh at Noon	38	36	31	26	23	22	22	26	32	36	38	39	
All-Sky Solar Radiation	RadAvg	2040	1744	1491	1211	961	759	889	1175	1563	1807	1915	2094	
	RadStd	156	160	133	104	66	68	68	69	91	168	178	188	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-4	-153	+331	+176

Nomenclature: See separate page